

Exploratory Data Analysis

To solve a business problem using analytics, you need to have historical data to come up with actionable insights. Data is the key - the better the data, the more insights you can get out of it. Typically, data comes from various sources, and your first job as a data analyst is to procure data from them. In this module, you have learnt about various sources of data and how to source data from public and private sources.

Data Sources and Formats

Public Data

A large amount of data collected by the government or other public agencies is made public for the purposes of research. Such data sets do not require special permission for access and are therefore called public data. Public data is available on the internet on various platforms. A lot of data sets are available for direct analysis, whereas some of the data must be manually extracted and converted into a format that is fit for analysis.

Advantages:

- Low acquisition cost
- Generally credible and authoritative

Limitations:

- May be outdated
- May lack business-specific detail
- Often inconsistent in format and quality

Private Data

Private data is that which is sensitive to organisations and is thus not available in the public domain. Banking, telecom, retail, and media are some of the key private sectors that rely heavily on data to make decisions. A large number of organisations seek to leverage data analytics to make crucial decisions. As organisations become customer-centric, they utilise insights from data to enhance customer experience, while also optimising their daily processes. There are various types of quality issues when it comes to data, and that's why data cleaning is one of the most time-consuming steps of data analysis. For example, there could be formatting errors (e.g. rows and columns are merged), missing values, repeated rows, spelling inconsistencies, etc.



These issues could make it difficult to analyse data and could lead to errors or irrelevant results. Thus, these issues need to be corrected before data is analysed.

Data Formats and Structure

Data formats describe how data is organised and stored. The structure of data affects storage choices, processing tools, and analytical techniques.

Based on structure, data can be classified into three broad categories.

Structured Data

Structured data follows a predefined schema with rows and columns.

ID	Name	Dept	Salary
E101	Asha Verma	Content	65000
E102	Rohan Mehta	HR	48000
E103	Sara Khan	Tech	90000
E104	Neeraj Rai	Marketing	52000

Features:

- Easy to store in relational databases
- Simple to query using SQL
- Suitable for reporting and dashboards

Examples include sales tables, customer master data, and inventory records.

Semi-Structured Data

Semi-structured data does not follow a rigid schema but contains tags or keys that provide partial organisation.

```
{ "title": "Data Pipelines",  
  "author": "J. Rao", "year": 2023,  
  "tags": ["ETL", "workflow"] }  
{ "title": "Python Patterns",  
  "author": "L. Menon", "year": 2021,  
  "tags": ["python", "design"] }  
{ "title": "Cloud Foundations",  
  "author": "A. Patel", "year": 2022,  
  "tags": ["AWS", "architecture"] }
```

Features:

- Flexible structure
- Common in APIs and web applications
- Stored in NoSQL databases or data lakes

Examples include JSON, XML, YAML files, and system logs.

Unstructured Data

Unstructured data lacks any predefined structure.

```
Data flowing inconsistently  
across regions.  
Maybe due to network  
routing.  
Needs further investigation  
tomorrow.
```

Features:

- Text-heavy or media-rich
- Difficult to analyse using traditional tools

- Requires advanced techniques such as NLP or computer vision

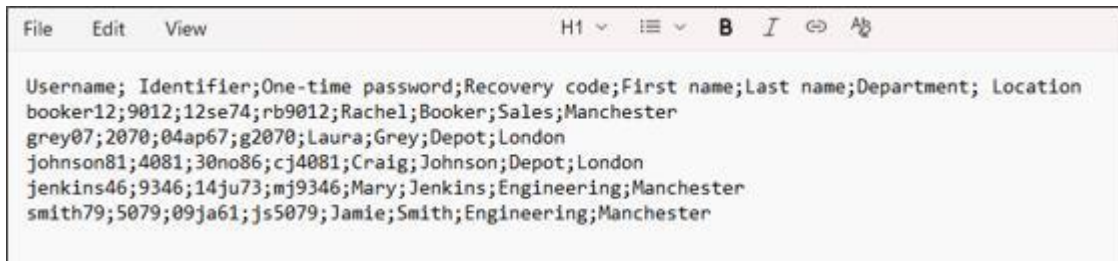
Examples include emails, PDFs, images, videos, audio recordings, and social media posts.

Common Tabular Data Formats

Tabular formats organise data into rows and columns, making them easy to read, store, clean, and analyse. They form the backbone of analytics workflows, databases, and machine learning pipelines.

CSV (Comma-Separated Values)

CSV is a plain-text format where each row represents a record, and columns are separated by commas.



A screenshot of a text editor window showing a CSV file. The window has a menu bar with 'File', 'Edit', and 'View'. The text content is as follows:

```
Username; Identifier;One-time password;Recovery code;First name;Last name;Department; Location
booker12;9012;12se74;rb9012;Rachel;Booker;Sales;Manchester
grey07;2070;04ap67;g2070;Laura;Grey;Depot;London
johnson81;4081;30no86;cj4081;Craig;Johnson;Depot;London
jenkins46;9346;14ju73;mj9346;Mary;Jenkins;Engineering;Manchester
smith79;5079;09ja61;js5079;Jamie;Smith;Engineering;Manchester
```

Key characteristics:

- Lightweight and human-readable
- Widely supported across tools and programming languages
- Stores only raw data without formatting

CSV files are commonly used for data exchange and initial analysis.

XLSX (Excel Spreadsheet)

XLSX is a spreadsheet format that supports advanced features for data manipulation and presentation.

	A	B	C	D
1	Course Name	Category	Duration (in weeks)	Fee (₹)
2	Data Structures	Computer Science	10	5000
3	Algorithms	Computer Science	8	4500
4	Web Development	Web Development	12	6000
5	Machine Learning	AI & ML	15	8000
6	Competitive Programming	Competitive Coding	20	7500
7				

Key characteristics:

- Supports multiple sheets
- Allows formulas, charts, and pivot tables
- Stores metadata such as styles and comments

Excel files are widely used in business and exploratory analysis.

TSV (Tab-Separated Values)

TSV is similar to CSV but uses tabs as delimiters instead of commas.

File	Edit	Format	View	Help
student_name	school_name	address	county	phone
Aleshia	Alan D Rosenberg	Cpa Pc	14 Taylor St	Kent 01944-36997
Evan	Cap Gemini America	5 Binney St	Buckinghamshire	01714-73768
France	"Elliott, John W Esq"	8 Moor Place	Bournemouth	01935-82166
Ulysses	"Mcmahan, Ben L"	505 Exeter Rd	Lincolnshire	01302-60130
Tyisha	Champagne Room	5396 Forth Street	West Midlands	01290-36728
Eric	"Thompson, Michael C Esq"	9472 Lind St	Northamptonshire	01545-81735
Marg	Wrangle Hill Auto Auct & Slvg	7457 Cowl St #70	Southampton	01362-62052
Laquita	In Communications Inc	20 Gloucester Pl #96	Tyne & Wear	01590-98248
Lura	Bizerba Usa Inc	929 Augustine St	South Gloucestershire	01340-71391
Yvette	Max Video	45 Bradfield St #166	Derbyshire	01933-51253

This format is particularly useful when data fields contain commas, such as addresses or textual descriptions.

JSON (JavaScript Object Notation)

JSON is a semi-structured format based on key-value pairs.

```
{
  "name": "Asha Verma",
  "month": "Jan",
  "sales": 1250,
  "items": [
    {"product": "Shoes", "quantity": 1},
    {"product": "Bag", "quantity": 1}
  ]
}
```

Key characteristics:

- Supports nested objects and arrays
- Widely used in APIs and modern data pipelines
- Easily handled in Python and JavaScript

Data Cleaning and Preprocessing

Real-world data is rarely available in a form that is immediately suitable for analysis. Before any meaningful insights can be extracted, raw data must be cleaned and preprocessed to ensure accuracy, consistency, and reliability. Data cleaning and preprocessing involve identifying and correcting errors, handling inconsistencies, and transforming raw data into a structured format that can be confidently used for analysis and modelling. Without these steps, analytical results may be misleading, biased, or entirely incorrect.

Need for Data Cleaning

Data cleaning is necessary because raw datasets commonly contain quality issues that can affect analysis of outcomes. Typical problems include:

- **Missing values**, where information is not recorded
- **Duplicate records**, often created during data collection or merging
- **Inconsistent formats**, such as mixed date or text representations
- **Incorrect or invalid values**, caused by entry errors or system limitations



If these issues are not addressed, analytical results may become misleading, biased, or incorrect. Data cleaning ensures that the dataset accurately reflects real-world conditions and supports dependable analysis.

Fixing Rows, Columns, and Data Types

A clean dataset must follow a tidy structure where each row represents a single observation; each column represents a single variable, and each variable has an appropriate data type.

Fixing Rows

Fixing rows ensures that every row corresponds to one valid data record. Common steps include:

- Removing summary rows such as totals or subtotals
- Deleting extra header or footer rows
- Removing blank rows
- Eliminating duplicate observations

Fixing Columns

Problem	Solution
'Customer Name '	<i>customer_name</i>
'Date of Order (DDMMYYYY)'	<i>order_date</i>
Two header rows	One header row
City + State combined	Split into separate columns or create standardised location
John Smith	Separated into <i>first_name</i> and <i>last_name</i>

Fixing columns ensures that each column captures exactly one variable. Common steps include:

- Merging columns to create unique identifiers
- Splitting combined columns to extract meaningful components
- Renaming columns consistently for clarity
- Removing irrelevant or unused columns
- Correcting misaligned data caused by improper imports

Fixing Data Types

Raw Value	Issue	Clean Value
'₹12,500'	Number as string	12500
'12/31/23'	Text date	2023-12-31
' Male '	Extra spaces	'Male'
'Gujarat' / 'gujarat' / 'GUJARAT'	Inconsistent casing	'Gujarat'

Fixing data types ensures that each column is stored in the correct format required for analysis and computation. Common steps include:

- Converting numeric values stored as strings into numeric types
- Parsing date and time values into standard datetime formats
- Ensuring categorical variables are represented consistently
- Correcting boolean values stored as text (e.g., "Yes"/"No", "True"/"False")

Correct data types are essential for accurate calculations, comparisons, and visualisations.

Handling Missing Values

Raw Value	Issue	Clean Value
Salary = ''	Missing numeric value	Impute with median / mean
City = ''	Missing categorical value	Fill with mode or 'Unknown'
85% missing values in <i>customer_comments</i>	Too many missing values	Drop column
Date = ''	Missing date	Forward fill / backward fill
Time series gaps such as 2024-01-01, 2024-01-04	Missing time points	Interpolate values for missing dates

Missing values are common in real-world data and may arise due to system failures, skipped inputs, or integration of issues across data sources. Missingness can be explicit (blank cells) or implicit (placeholder values such as "NA", "999", or "unknown").

Common strategies for handling missing values include:

- Identifying and standardising implicit missing value representations
- Deleting rows or columns when missing values are insignificant or excessively high
- Filling missing values using business judgement or statistical methods
- Retaining missing values when imputation may introduce bias

The chosen approach depends on the extent of missingness and its impact on analysis validity.

Handling Outliers

Outliers are observations that lie far from the majority of data points and can significantly affect analysis results.



Outliers can:

- Distort averages and summary statistics
- Create misleading patterns or trends
- Reduce the performance of statistical and machine learning models

Outliers may represent genuine rare events or data errors. Each outlier should be examined carefully before deciding whether to retain, correct, transform, or remove it.

Standardising Values

Standardisation ensures uniformity across the dataset and improves comparability between observations.

This process typically includes:

- Converting measurements to a common unit or scale
- Aligning different measurement systems
- Standardising numeric precision
- Unifying category labels with consistent naming conventions

Standardised data reduces ambiguity and ensures that inconsistencies do not influence analytical outcomes.

Fixing Invalid Values

Invalid values occur when data violates expected formats, types, or logical rules.

Common examples include:

- Values outside defined permissible ranges
- Logical inconsistencies across related variables
- Violations of internal business rules or constraints

Such values may arise from encoding issues, incorrect data entry, or flawed collection processes. Identifying and correcting or removing invalid values is essential for maintaining data integrity.



Filtering Data

Filtering focuses on the dataset on relevant and meaningful records, reducing noise, and improving analysis quality.

Common filtering operations include:

- Deduplicating records
- Filtering data based on date ranges, geography, or customer segments
- Retaining only variables relevant to the analysis objective

Effective filtering ensures that the dataset remains aligned with the analytical goal and improves computational efficiency.

Exploratory Data Analysis

Exploratory data analysis (EDA) is the first and foremost step to analyse any kind of data. Rather than a specific set of procedures, EDA is an approach, or a philosophy, which seeks to explore the most important and often hidden patterns in a data set. In EDA, we explore the data and try to come up with a hypothesis about it which we can later test using hypothesis testing. Statisticians use it to take a bird's eye view of the data and try to make some sense of it.

Types of Exploratory Analysis

EDA can be broadly classified based on the number of variables analysed and whether segmentation is involved. The main types include:

- **Univariate Analysis**, which focuses on a single variable
- **Segmented Univariate Analysis**, which examines a single variable across different categories
- **Bivariate Analysis**, which studies the relationship between two variables
- **Segmented Bivariate Analysis**, which explores relationships between two variables across subgroups

Each type serves a distinct purpose and provides complementary insights into the dataset.

Univariate Analysis

Univariate analysis involves examining one variable at a time to understand its distribution, central tendency, and variability. This form of analysis helps build a foundational understanding of individual variables before exploring interactions between them.

Univariate analysis is commonly used to:

- Identify common or dominant values
- Detect skewness and outliers
- Understand the overall behaviour and range of a variable

The analytical approach depends on the type of variable being studied:

- **Nominal variables** are analysed using frequency tables and bar charts to understand category proportions
- **Ordinal variables** are analysed using ordered frequency plots that respect category ranking
- **Quantitative variables** are analysed using histograms, box plots, and summary statistics such as mean, median, and standard deviation

Segmented Univariate Analysis

Segmented univariate analysis extends univariate analysis by examining a numerical variable across different categories or segments. Instead of analysing the variable in isolation, it compares how the variable behaves within each group.

The typical steps involved include:

- Grouping the data by a categorical variable
- Computing summary statistics for each group
- Comparing distributions or metrics across groups

This approach helps uncover group-specific patterns that may be hidden when analysing aggregated data, enabling more nuanced insights.

Bivariate Analysis

Bivariate analysis focuses on understanding the relationship between two variables. It helps determine whether changes in one variable are associated with changes in another and provides insight into the direction and strength of such relationships.

Depending on the variable types involved, bivariate analysis may include:

- **Numerical–numerical analysis**, using scatter plots and correlation measures
- **Categorical–categorical analysis**, using cross tables or contingency tables
- **Numerical–categorical analysis**, using grouped visualisations such as box plots or bar charts

Bivariate analysis is essential for identifying associations that may inform feature selection and model design.

Segmented Bivariate Analysis

Segmented bivariate analysis examines the relationship between two numerical variables across different levels of a third categorical variable. This allows analysts to assess whether observed relationships remain consistent or vary across subgroups.

Common segmentation variables include:

- Gender
- Region
- Customer type
- Product category

By incorporating segmentation, analysts can detect interaction effects and subgroup-specific trends that may be masked in overall bivariate analysis.

Derived Metrics

There are three different types of derived metrics:

- Type-driven metrics
- Business-driven metrics
- Data-driven metrics

Type-driven metrics

These metrics can be derived by understanding the variable's typology. You have already learnt one simple way of classifying variables/attributes — categorical (ordered, unordered) and quantitative or numeric. Similarly, there are various other ways of classification, one of which is Steven's typology.

Steven's typology classifies variables into four types - nominal, ordinal, interval and ratio:

- **Nominal variables:** Categorical variables, where the categories differ only by their names; there is no order among categories, e.g. colour (red, blue, green), gender (male, female), department (HR, analytics, sales). These are the most basic forms of categorical variables.
- **Ordinal variables:** Categories follow a certain order, but the mathematical difference between categories is not meaningful, e.g. education level (primary school, high school, college), height (high, medium, low), performance (bad, good, excellent), etc. Ordinal variables are nominal as well
- **Interval variables:** Categories follow a certain order, and the mathematical difference between categories is meaningful, e.g. temperature in degrees celsius (the difference between 40 and 30 degrees C is meaningful), dates (the difference between two dates is the number of days between them), etc. Interval variables are both nominal and ordinal
- **Ratio variables:** Apart from the mathematical difference, the ratio (division/multiplication) is possible, e.g. sales in dollars (\$100 is twice \$50), marks of students (50 is half of 100), etc. Ratio variables are nominal, ordinal and interval type

Understanding types of variables enables you to derive new metrics of types different from the same column. For example, age in years is a ratio attributes, but you can convert it into an ordinal type by binning it into categories such as children (< 13 years), teenagers (13-19 years), young adults (20-25 years), etc. This enables you to ask questions, e.g. do teenagers do X better than children, are young adults more likely to do X than the other two types, etc. Here, X is an action you are interested in finding.

Business Driven Metrics:

They are derived from the existing variables, but it requires domain expertise. Driving metrics from a business perspective is not an easy task. Without understanding the domain correctly, deriving insights becomes difficult and prone to errors.

Data Driven Metrics:

Data-driven metrics can be created based on the variables present in the existing data set. For example, if you have two variables in your data set, such as "weight" and "height" which shows a high correlation. So, instead of analysing "weight" and "height" variables separately, you can think of deriving a new metric "Body Mass Index (BMI)". Once you get the BMI, you can easily categorise people based on their fitness, e.g. a BMI below 18.5 should be considered an underweight category, while BMI above 30.0 is considered obese, by standard norms. This is how data-driven metrics can help you discover hidden patterns out of the data.

Exploratory data analysis helps you, as a data analyst, look beyond the data. It is a never-ending process — the more you explore the data, the more insights you get. Almost 80% of the time, you would spend your time as a data analyst understanding the data and solving various business problems through EDA. If you understand EDA properly, then half the battle is won.

So far in this module, you have learnt the five most crucial topics for any kind of analysis. They are as follows:

- Understanding domain
- Understanding data and preparing it for analysis
- Univariate analysis and segmented univariate analysis
- Bivariate analysis
- Deriving new metrics from the existing data

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